

Eugene Lafayette Buchanan

Senior Software Engineer & Technical Consultant

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Professional Summary

Senior software engineer with 25+ years building, shipping, implementing and supporting solutions. Expert quality assurance (QA), test automation integrations for ERP, education and streaming platforms. I also work as a renewable energy consultant and real estate professional, so I move easily between technical depth and business operations. I write clean code, build reliable test systems, and explain complex ideas clearly.

Skills

Languages & Frameworks: JavaScript, TypeScript, Python, Rust, PHP, Node.js, React, SQL

Testing & QA: Cypress, Selenium, Postman, JUnit, Qase, white-box testing, regression suites, performance and load testing, accessibility testing, security testing

Cloud, CI/CD & DevOps: AWS (ECS, S3), CI/CD pipelines, Docker, Git, Bitbucket, Elasticsearch, Unix domain sockets, JSON-RPC

Integrations & Auth: SAP, Oracle, Workday, Coursera, UKG, Zoom, SAML, OAuth, SFTP, REST APIs

AI & Automation: AI automation, prompt engineering, large language model (LLM) integration, agentic AI, LangGraph, Model Context Protocol (MCP)

Platforms & Domains: ERP, CRM, learning management systems (LMS), streaming media, enterprise software, alternative energy, real estate

Leadership & Communication: Team building, mentoring, training, cross-functional collaboration, product ownership, backlog management, presales engineering, technical writing

Work Experience — Software Engineering

Senior QA Engineer / Test Automation Lead / Product Owner — Integrations

Knowledgecity LLC · Remote · Dec 2020 – Aug 2025

- Led QA across integration, frontend, backend, mobile, security, performance, load, API, database, functionality, usability, accessibility, and localization testing. Reduced technical debt.
- Built and maintained a CI/CD quality pipeline with Cypress, Selenium, Postman, JUnit, Elasticsearch, and S3 on AWS.
- Hired and trained the test automation team. Built regression suites, health dashboards, and alert systems for complex deployments.
- Shipped integrations with SAP, Oracle, Workday, Coursera, UKG, and Zoom using SAML, OAuth, SFTP, and custom REST APIs.
- Managed dev and test teams, the product backlog, and design reviews as Integrations Product Owner.
- Applied AI prompt engineering and shipped chatbot support tooling to production.

Senior Software Engineer II

RealNetworks · Seattle, WA · 1999 – 2001

- Ran QA on an advanced research team covering consumer appliances, mobile platforms, stream servers, and cellular networks.
- Led global presales engineering for consumer devices and trained new hires.

Earlier Software Test Engineering Roles

Microsoft · IBM · Keene Inc.

- Microsoft, Redmond, WA (1999): Senior Software Test Engineer IV, Windows Media Server; white-box testing and development.
- IBM, Kirkland, WA (1998): Senior Test Engineer III, high-availability servers.
- Microsoft, Redmond, WA (1997): Software Test Engineer II, Windows 98 and Windows NT.
- Keene Inc., Seattle, WA (1996): Technical Support Agent II, Windows 95 and Windows NT.

IT Manager

Lucerne Valley Unified School District · Lucerne Valley, CA · 2002

- Maintained IT systems across four school campuses, including WAN/LAN, NetWare servers, virtualized desktops, and audio/video equipment.
- Met state attendance reporting requirements.

Freelance IT Consultant

Amp Computer · Apple Valley, CA · 2003 – 2011

- Sold and supported servers and PCs. Configured ERP and CRM systems, office automation, and custom solutions.

Work Experience — Additional Professional Roles

Alternative Energy Consultant

2004 – Present

- PhaseSave.com, Palm Springs, CA (2017 – Present): Director of product development, sales engineering, project management, and sourcing. Expertise in solar power, air conditioning, phase-change materials, and cold chain logistics.
- EWSolar.net, Homeland, CA (2014 – 2016): Vice President, large-scale solar-powered water pumping projects.
- Free Energy Resources Inc., Phelan, CA (2013): Chief Technology Officer, solar marketing; sales, procurement, and project management.
- GRIDNOT Inc., Lucerne Valley, CA (2007 – 2012): Vice President, sales and integration of solar, wind, and geothermal air-conditioning systems.
- OnPoint Power Inc. (2006), Desert Power Inc. (2005), Solatron Inc. (2004): renewable energy sales and support.

Business Consultant

2013 – 2016

- Medico Investments LLC, Hesperia, CA: recovered a non-permitted photovoltaic installation and worked with the city for permission to operate using as-built engineering.
- Foremost Organization, Hesperia, CA: built a nonprofit and community outreach program; filed federal, state, and city applications, and ran a community social program.
- Provided campus IT support and security and backup policies; upgraded video surveillance and installed campus-wide wireless.
- Reviewed business plans and cut procurement, operations, and staffing costs. Won city approval for a bridge to landlocked land.

Real Estate Professional & Elected Board President

1999 – Present

- Tierrachain.com (2023 – Present): founder of a land technology company building software as a service (SaaS) that uses AI and blockchain to improve real estate exchange.
- Mariana Ranchos County Water Board (2006 – 2010): elected board president; youngest in California history. Cut budget waste by 53%, replaced corrupt personnel, hired a strong general manager, and blocked a high-risk floodplain project that later flooded.

Open-Source Contributions

Open-Source Contributor — Agentic AI Infrastructure

ZeroClaw Labs · github.com/zeroclaw-labs/zeroclaw · 32.2k GitHub stars · 2026 – Present

- Designed and shipped an observer that reports agent lifecycle states over JSON-RPC on Unix domain sockets. Resolved four rounds of review feedback, including bounded I/O to prevent hangs and env-driven discovery.
- Reviewed and contributed architecture feedback on a multi-user authentication feature: permission profiles, principal isolation, and a workspace multi-tenant axis.

Education

- Associate Degree, Victor Valley College
- California Notary Commission
- Toastmasters International
- High School Diploma, Lucerne Valley High School